Name: Chelsea Stacy

Date: 6/1/2026

Course: FDN 130

Assignment 7- Functions

Introduction

This is a document intended to demonstrate knowledge gained from module 7 of this course on functions.

Using a SQL UDF

A SQL user defined function (or UDF) is a custom function that can be used when there is not a SQL Server built-in function that can meet the user's needs. A UDF can return a scalar value as an expression or a table of values.

One example of a use case for a scalar UDF is when a user would like to create a Check constraint in order to restrict data entry. A table-valued UDF might be used in instances where a developer would like to filter a view by a specific parameter.

Differences Between a Scalar, In-Line, and Multi-Statement Functions

There are a few different types of functions that serve different purposes.

Scalar Function

A scalar function is one that returns a single value as an expression. It can take parameters and can be applied to each row of a result set.

In-Line Function

An in-line function is one that produces an entire table as output, instead of just a single expression. It uses a single SELECT statement and parameters to produce a table. The table's structure is determined by the chosen SELECT statement.

Multi-Statement Function

A multi-statement function is one that uses more than one statement to produce a table of values. In this type of function, the structure of each column in the table must be declared, including data type.

Summary

Functions are a useful tool when writing SQL queries. While SQL Server has premade functions that can be used, a user is also able to write their own functions, known as 'user-defined functions.’ A function can be scalar (meaning it produces a single value) or it can produce an entire table. In-line functions can produce a simple table based off of a single SELECT statement, while a multi-statement function adds a layer of complexity, allowing for multiple statements to produce a table that the user defines for themselves.